

Asymptotic Normality

- Assumption R5: The pdf $f(x; \theta)$ is three times differentiable as a function of θ . Further, for all $\theta \in \Omega$, there exists a constant c and a function $M(x)$ such that

$$\left| \frac{\partial^3}{\partial \theta^3} \log f(x; \theta) \right| \leq M(x),$$

with $\mathbb{E}_{\theta_0} M(x) < \infty$ for all $\theta_0 - c < \theta < \theta_0 + c$ and all x in the support of X .

- Theorem 6.2.2.: Assume $X_1, \dots, X_n$ are iid with pdf $f(x, \theta_0)$ such that R0-R5 are satisfied. $\hat{\theta}$ is the MLE. Then,

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{D} N(0, \frac{1}{I(\theta_0)}).$$

- **Definition:** Let $X_1, \dots, X_n$ be iid with pdf $f(x; \theta_0)$. Suppose $\hat{\theta}_{1n} = \hat{\theta}_{1n}(X_1, \dots, X_n)$ is an estimator of θ_0 such that $\sqrt{n}(\hat{\theta}_{1n} - \theta_0) \xrightarrow{D} N(0, \sigma_{\hat{\theta}_{1n}}^2)$. Then

- The **asymptotic efficiency** of $\hat{\theta}_{1n}$ is defined as

$$e(\hat{\theta}_{1n}) = \frac{1/I(\theta_0)}{\sigma_{\hat{\theta}_{1n}}^2}.$$

- The estimator $\hat{\theta}_{1n}$ is said to be **asymptotically efficient** if the ratio $e(\hat{\theta}_{1n})$ is 1.
- Let $\hat{\theta}_{2n}$ be another estimator such that $\sqrt{n}(\hat{\theta}_{2n} - \theta_0) \xrightarrow{D} N(0, \sigma_{\hat{\theta}_{2n}}^2)$. The the **ARE** of $\hat{\theta}_{1n}$ to $\hat{\theta}_{2n}$ is the reciprocal of the ratio of their respective asymptotic variances, i.e.,

$$e(\hat{\theta}_{1n}, \hat{\theta}_{2n}) = \frac{\sigma_{\hat{\theta}_{2n}}^2}{\sigma_{\hat{\theta}_{1n}}^2}.$$

- **ARE of the sample Median to the sample Mean:** We will obtain this ARE under the Laplace and normal distributions. Consider first the Laplace model,

$$X_i = \theta + e_i, \quad i = 1, \dots, n,$$

where the e_i s are iid with Laplace pdf $f(x; \theta) = \frac{1}{2}e^{-|x|}$. We know that the mle is the median Q_2 and $I(\theta_0) = 1$. Hence $\sqrt{n}(Q_2 - \theta_0) \xrightarrow{D} N(0, 1)$. By CLT, the sample mean $\bar{X}$ is asymptotic normal with mean θ_0 and variance σ^2/n and $\sigma^2 = \text{Var}(X_i) = 2$. So, $\text{ARE}(Q_2, \bar{X}) = 2$.

Now suppose that pdf of e_i is $N(0, 1)$. We will show in Chapter 10 that Q_2 is asymptotically normal with mean θ_0 and variance $(\pi/2)/n$. Because that variance of $\bar{X}$ is $1/n$, so $\text{ARE}(Q_2, \bar{X}) = 2/\pi = 0.636$.

- Since $I(\hat{\theta}_n) \xrightarrow{P} I(\theta_0)$ and $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{D} N(0, I^{-1}(\theta_0))$, the following interval is an approximate $(1 - \alpha)100\%$ confidence interval for θ ,

$$\left(\hat{\theta}_n - z_{\alpha/2} \frac{1}{\sqrt{nI(\hat{\theta}_n)}}, \hat{\theta}_n + z_{\alpha/2} \frac{1}{\sqrt{nI(\hat{\theta}_n)}} \right)$$

- Corollary 6.2.2.: Suppose $g(x)$ is continuous function of x which is differentiable at θ_0 such that $g'(\theta_0) \neq 0$ Then,

$$\sqrt{n}(g(\hat{\theta}_n) - g(\theta_0)) \rightarrow^D N(0, \frac{g'(\theta_0)^2}{I(\theta_0)}).$$

- Let $\bar{X}$ be the mean of a random sample of size n from a $N(\theta, \sigma^2)$ distribution. Assume $\sigma^2 = 1$ is known. Show that $\bar{X}^2 - \frac{1}{n}$ is an unbiased estimator of θ^2 and find its efficiency.

